

Appendix D - Per-Engine MMIO Maps

Every memory-mapped device, in address order. Each register is 32-bit on the bus unless noted; only the documented low bits carry information.

D.1 Video chip (0xF0000-0xF049B)

Address	Name	R/W	Notes
0xF0000	VIDEO_CTRL	R/W	Enable, mode select, sync source.
0xF0004	VIDEO_MODE	R/W	Mode encoding: 320x200, 320x240, 640x480, 800x600, 1024x768, 1280x960, 1920x1080.
0xF0008	VIDEO_STATUS	R	VBlank flag, HBlank flag, raster line.
0xF000C	VIDEO_FB_BASE	R/W	Framebuffer base address.
0xF0010	COPPER_CTRL	R/W	Copper enable.
0xF0014	COPPER_PTR	R/W	Copper-list base address.
0xF0018	COPPER_PC	R	Current copper program counter.
0xF001C	COPPER_STATUS	R	Copper running / stopped.
0xF0020-0xF0044	BLT_*	R/W	Blitter source A/B/C/D, mask, modulo, control.
0xF0048-0xF0054	VIDEO_RASTER_*	R/W	Raster split lines and per-line palette swaps.
0xF0078-0xF0477	palette	R/W	256-entry 32-bit RGBA palette.
0xF0488-0xF049B	BLT_EXT_*	R/W	Extended blitter (large modes).

D.2 Terminal / serial / input (0xF0700-0xF07FF)

Address	Name	R/W	Notes
0xF0700	TERM_OUT	W	Print one byte.
0xF0704	TERM_STATUS	R	Bit 0 input avail, bit 1 output ready.
0xF0708	TERM_IN	R	Dequeue cooked input byte.
0xF070C	TERM_LINE_STATUS	R	Bit 0 complete line in queue.
0xF0710	TERM_ECHO	R/W	Local echo.
0xF0724	TERM_CTRL	R/W	Bit 0 line-input mode.
0xF0728	TERM_KEY_IN	R	Dequeue raw key.
0xF072C	TERM_KEY_STATUS	R	Bit 0 raw key avail.
0xF0730	MOUSE_X	R	Absolute X (low 16 bits).
0xF0734	MOUSE_Y	R	Absolute Y.
0xF0738	MOUSE_BUTTONS	R	Bit 0 L, 1 R, 2 M.

Address	Name	R/W	Notes
0xF073C	MOUSE_STATUS	R	Bit 0 changed (clears on read).
0xF0740	SCAN_CODE	R	Dequeue raw scancode.
0xF0744	SCAN_STATUS	R	Bit 0 scancode avail.
0xF0748	SCAN_MODIFIERS	R	Bit 0 shift, 1 ctrl, 2 alt, 3 caps.
0xF074C	MOUSE_CTRL	R/W	Bit 0 request relative mode.
0xF0750	RTC_EPOCH	R	Unix epoch seconds.
0xF0754	MOUSE_DX	R	Signed accumulated dX (clears on read).
0xF0758	MOUSE_DY	R	Signed accumulated dY.
0xF07F0	TERM_SENTINEL	W	Write 0xDEAD to stop CPU.

D.3 SoundChip (0xF0800-0xF0B7F)

Range	Block	Notes
0xF0800-0xF08FF	Global	AUDIO_CTRL, ENV_SHAPE, filter, reverb, overdrive.
0xF0900-0xF093F	Square	SQ_FREQ, SQ_DUTY, SQ_ENV_*, SQ_GATE.
0xF0940-0xF097F	Triangle	Same shape.
0xF0980-0xF09BF	Sine	Same shape.
0xF09C0-0xF09FF	Noise	LFSR period, volume, gate.
0xF0A00-0xF0A1F	Sync / ring-mod	Source-channel selects.
0xF0A20-0xF0A6F	Sawtooth	Same shape.
0xF0A80-0xF0B7F	Flex 0-3	64-byte per-channel block.

D.4 SFX triggers (0xF0E80-0xF0EFF)

Per-channel offset	Field	Notes
+0x00	SFX_PTR	Sample base address.
+0x04	SFX_LEN	Sample length in bytes.
+0x08	SFX_LOOP_PTR	Loop point.
+0x0C	SFX_LOOP_LEN	Loop length.
+0x10	SFX_FREQ	Playback rate.
+0x14	SFX_VOL	Volume (0-64).
+0x18	SFX_FORMAT	Sample format.
+0x1C	SFX_CTRL	Trigger / one-shot / loop.

Four channels at 0xF0E80, 0xF0EA0, 0xF0EC0, 0xF0EE0.

D.5 PSG / AY-3-8910 (0xF0C00-0xF0C20)

Offset	Register
+0x00-+0x0F	R0-R15, one byte per register (byte-contiguous).
+0x10	PSG_PLAY_PTR (32-bit).
+0x14	PSG_PLAY_LEN (32-bit).
+0x18	PSG_PLAY_CTRL (bit 0 start, bit 1 stop, bit 2 loop).
+0x1C	PSG_PLAY_STATUS (bit 0 busy, bit 1 error).
+0x20	PSG_PLUS_CTRL.

D.6 SN76489 (0xF0C30-0xF0C37)

Documented in full in Chapter 14.

D.7 SID family

Each SID instance is a byte-contiguous block in the original 6581/8580 register layout. Three independent instances live at three bases:

Block	Range	Notes
SID	0xF0E00-0xF0E1C	Primary SID.
SID2	0xF0E30-0xF0E4C	Second independent SID.
SID3	0xF0E50-0xF0E6C	Third independent SID.

Within each instance the offsets follow the original chip layout, one byte per register:

Offset	Register
+0x00-+0x06	Voice 1: FREQ_LO, FREQ_HI, PW_LO, PW_HI, CTRL, AD, SR.
+0x07-+0x0D	Voice 2 (same shape).
+0x0E-+0x14	Voice 3 (same shape).
+0x15-+0x17	Filter: FC_LO, FC_HI, RES_FILT.
+0x18	MODE_VOL.
+0x19	SID+ control (SID_PLUS_CTRL) / POT_X mirror.
+0x1A	POT_Y.
+0x1B	OSC3.
+0x1C	ENV3.

The primary SID block at 0xF0E00 has an attached .sid file player block in the same 32-byte window:

Offset (from 0xF0E00)	Register
+0x20	SID_PLAY_PTR (32-bit).

Offset (from 0xF0E00)	Register
+0x24	SID_PLAY_LEN (32-bit).
+0x28	SID_PLAY_CTRL (start / stop / loop).
+0x2C	SID_PLAY_STATUS (busy / error).
+0x2D	SID_SUBSONG.

The two flexible audio blocks 0xF0C40-0xF0CFF and 0xF0D40-0xF0DFF are not SID2 and SID3. They are extra-channel mirrors (channels 4-6 and 7-9 in the global SoundChip channel space) and have the flex-channel layout documented under SoundChip (D.3), not the SID layout. A program that wants SID2 / SID3 should program them at 0xF0E30 and 0xF0E50.

D.8 POKEY (0xF0D00-0xF0D20)

Offset	Register
+0x00	POKEY_AUDF1.
+0x04	POKEY_AUDC1.
+0x08-+0x1C	AUDF2-AUDF4, AUDC2-AUDC4.
+0x20	POKEY_AUDCTL.
+0x24-+0x2F	SAP_PLAY_* player.

D.9 TED audio (0xF0F00-0xF0F1F)

Offset	Register
+0x00	TED_FREQ1_L0.
+0x04	TED_FREQ1_HI.
+0x08	TED_FREQ2_L0.
+0x0C	TED_FREQ2_HI.
+0x10	TED_SND_CTRL.
+0x14-+0x1F	TED_PLAY_* player.

D.10 TED video (0xF0F20-0xF0F5F)

40x25 text with 121-colour cells; see Chapter 6.

D.11 VGA (0xF1000-0xF13FF)

Range	Subsystem
0xF1000-0xF103F	VGA_MODE, VGA_STATUS, VGA_CTRL.
0xF1040-0xF107F	Sequencer (VGA_SEQ_*).
0xF1080-0xF10FF	CRTC (VGA_CRTC_*).

Range	Subsystem
0xF1100-0xF113F	Graphics controller (VGA_GC_*).
0xF1140-0xF117F	Attribute controller (VGA_ATTR_*).
0xF1180-0xF11FF	DAC + palette index/data.
0xF1200-0xF13FF	Palette RAM.

D.12 ULA (0xF2000-0xF2017)

Offset	Register
+0x00	ULA_BORDER (bits 0-2).
+0x04	ULA_CTRL.
+0x08	ULA_STATUS.
+0x0C	ULA_ADDR_LO.
+0x10	ULA_ADDR_HI.
+0x14	ULA_DATA.

ULA VRAM aperture at 0xFA000-0xFBFFF.

D.13 ANTIC + GTIA (0xF2100-0xF21B7)

ANTIC block at 0xF2100-0xF213F with DMACTL, CHACTL, DLISTL/H, HSCROL, VSCROL, PMBASE, CHBASE, WSYNC, VCOUNT, NMEN/IST/RES. GTIA at 0xF2140-0xF21B7 with COLPM0-3, COLPF0-3, COLBK, PRIOR, GRCTL, CONSOL.

D.14 File I/O (0xF2200-0xF221F)

Offset	Register
+0x00	FILE_NAME_PTR.
+0x04	FILE_DATA_PTR.
+0x08	FILE_DATA_LEN.
+0x0C	FILE_CTRL (write triggers).
+0x10	FILE_STATUS.
+0x14	FILE_RESULT_LEN.
+0x18	FILE_ERROR_CODE.

D.15 Amiga Paula DMA (0xF2260-0xF22AF)

Four-channel DMA sample engine. Per-channel address, length, period, volume, and control registers. Documented in Chapter 21.

D.16 Media loader (0xF2300-0xF231F)

Dispatches by file extension into the matching audio engine. Used by SOUND PLAY (Chapter 22).

D.17 Image executor (0xF2320-0xF233F)

The RUN keyword writes the image filename here and triggers the load. The block has filename pointer, control, status, and result fields shaped like the File I/O block.

D.18 Coprocessor (0xF2340-0xF238F)

COSTART, COSTOP, COCALL, COSTATUS, COWAIT, plus the ring-buffer base/head/tail registers. Six worker types (Chapter 31).

D.19 IRQ diagnostics (0xF23C0-0xF23DF)

Offset	Register
+0x00	In-service mask.
+0x04	State flags.
+0x08	Pending mask.
+0x0C	Delivered counters.
+0x10	Blocked counters.
+0x14	RTE count.
+0x18	Consecutive STOP spins.
+0x1C	Watchdog event count.

D.20 SysInfo (0xF2400-0xF24FF)

Offset	Register
+0x00	Total RAM, low 32 bits.
+0x04	Total RAM, high 32 bits.
+0x08	Active RAM, low.
+0x0C	Active RAM, high.

D.21 HOST appliance block (0xF1400-0xF140F)

Offset	Name	Width	Notes
+0x00	command	byte	Subverb enum (W).
+0x04	trigger	byte	Non-zero fires the command (W).
+0x08	status	byte	Terminal-state enum (R).

Offset	Name	Width	Notes
+0x0C	exit	32-bit	Full 32-bit exit code from the underlying action (R); byte lanes at +0x0C-+0x0F return successive bytes of the same value.

See Chapter 35 for the subverb enum and the state machine. Reachable from IE64, IE32, M68K, and x86; not reachable from the 6502 or Z80.

D.22 Voodoo 3D (0xF8000-0xF87FF)

Status, framebuffer base, clip rect, triangle setup, texture descriptors, fog, alpha, chroma-key, Z-buffer. Documented in Chapter 9. Texture RAM at 0xD0000-0xDFFFF.